Simboli e loro proprietà

Simbolo	Nome	Proprietà
$\lvert A\rvert$	cardinalità dell'insieme A	$\lvert A\cup B\rvert = \lvert A\rvert + \lvert B\rvert$ se $A\cap B=\emptyset$
$\mathcal{P}(\Omega)$	insieme delle parti dell'insieme Ω	$\big\lvert\mathcal{P}(\Omega)\big\rvert = 2^{\lvert\Omega\rvert}$
$\mathbb{1}_A(x) := \begin{cases} 0 & \text{se } x\notin A \\ 1 & \text{se } x\in A \end{cases}$	funzione indicatrice dell'insieme A	$\mathbb{1}_A\cdot\mathbb{1}_B = \mathbb{1}_{A\cap B}$
$\binom{n}{k} := \begin{cases} \dfrac{n!}{k!(n-k)!} & \text{se } k\in\{0,\dots,n\} \\ 0 & \text{altrimenti} \end{cases}$	coefficiente binomiale di n su k	$\binom{n}{k-1}+\binom{n}{k}=\binom{n+1}{k}$
$\Gamma(\alpha) := \displaystyle\int_0^{+\infty} x^{\alpha-1}\mathrm{e}^{-x}\mathrm{d}x \quad (\text{con } \alpha>0)$	funzione gamma di Eulero	$\Gamma(\alpha+1)=\alpha\Gamma(\alpha)$ $\Gamma(n+1)=n! \quad \text{se } n\in\mathbb{N}$ $\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$

Regole di conteggio

Definizione	Nome	Formula
$\big\lvert\{f : A\to B\}\big\rvert$	disposizioni con ripetizione di $\lvert B\rvert$ oggetti in classi di $\lvert A\rvert$	$\lvert B\rvert^{\lvert A\rvert}$
$\big\lvert\{f : A\to B \mid f \text{ è iniettiva}\}\big\rvert$ $(\text{con } \lvert B\rvert\geq\lvert A\rvert)$	disposizioni senza ripetizione di $\lvert B\rvert$ oggetti in classi di $\lvert A\rvert$	$\dfrac{\lvert B\rvert!}{(\lvert B\rvert-\lvert A\rvert)!}$
$\big\lvert\{A\subseteq\Omega \mid \lvert A\rvert=k\}\big\rvert$	combinazioni senza ripetizione di $\lvert\Omega\rvert$ oggetti in classi di k	$\binom{\lvert\Omega\rvert}{k}$

Densità di variabili aleatorie discrete

$X\sim$	Parametri	$p_X(k)$	k	$\mathbb{E}[X]$	$\mathrm{Var}[X]$	$m_X(t)$
$B(1,q)$ (bernoulliana)	$q\in(0,1)$	$q^k(1-q)^{1-k}$	$0,1$	q	$q(1-q)$	$q(\mathrm{e}^t-1)+1$
$B(n,q)$ (binomiale)	$n\in\{1,2,\dots\}$ $q\in(0,1)$	$\binom{n}{k}q^k(1-q)^{n-k}$	$0,\dots,n$	nq	$nq(1-q)$	$\big[q(\mathrm{e}^t-1)+1\big]^n$
$\mathcal{H}(r+b,r,n)$ (ipergeometrica)	$b,r\in\{1,2,\dots\}$ $n\in\{1,\dots,b+r\}$	$\dfrac{\binom{r}{k}\binom{b}{n-k}}{\binom{b+r}{n}}$	$\max\{0,n-b\},$ $\dots,\min\{n,r\}$	$\dfrac{nr}{b+r}$	$\dfrac{nbr}{(b+r)^2}\times$ $\times\left(1-\dfrac{n-1}{b+r-1}\right)$	–
$\mathcal{G}(q)$ (geometrica)	$q\in(0,1)$	$q(1-q)^{k-1}$	$1,2,\dots$	$\dfrac{1}{q}$	$\dfrac{1-q}{q^2}$	$\dfrac{q\mathrm{e}^t}{1-(1-q)\mathrm{e}^t}$ per $t<-\ln(1-q)$
$\mathcal{P}(\lambda)$ (poissoniana)	$\lambda>0$	$\mathrm{e}^{-\lambda}\dfrac{\lambda^k}{k!}$	$0,1,\dots$	λ	λ	$\exp\big(\lambda(\mathrm{e}^t-1)\big)$
$\mathcal{U}(\{a,\dots,b\})$ (uniforme discr.)	$a,b\in\mathbb{Z}$ t.c. $a<b$	$\dfrac{1}{b-a+1}$	$a,\dots,b$	$\dfrac{a+b}{2}$	$\dfrac{(b-a+1)^2-1}{12}$	$\dfrac{\mathrm{e}^{(b+1)t}-\mathrm{e}^{at}}{(b-a+1)(\mathrm{e}^t-1)}$

Densità di variabili aleatorie assolutamente continue

$X \sim$	Parametri	$f_X(z)$	$\mathbb{E}[X]$	$\mathrm{Var}[X]$	$m_X(t)$
$\mathcal{U}([a,b])$ (uniforme cont.)	$a, b \in \mathbb{R}$ t.c. $a < b$	$\frac{1}{b-a}\,\mathbb{1}_{[a,b]}(z)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{\mathrm{e}^{bt}-\mathrm{e}^{at}}{(b-a)t}$
$\mathcal{E}(\lambda)$ (esponenziale)	$\lambda > 0$	$\lambda\,\mathrm{e}^{-\lambda z}\,\mathbb{1}_{[0,+\infty)}(z)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda - t}$ per $t < \lambda$
$\Gamma(\alpha,\lambda)$ (gamma)	$\alpha, \lambda > 0$	$\frac{\lambda^\alpha}{\Gamma(\alpha)}\,z^{\alpha-1}\,\mathrm{e}^{-\lambda z}\,\mathbb{1}_{[0,+\infty)}(z)$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$	$\left(\frac{\lambda}{\lambda - t}\right)^\alpha$ per $t < \lambda$
$\chi^2(m)$ (chi-quadrato)	$m \in \{1, 2, \ldots\}$	$\frac{1}{2^{\frac{m}{2}}\Gamma\left(\frac{m}{2}\right)}\,z^{\frac{m}{2}-1}\,\mathrm{e}^{-\frac{z}{2}}\,\mathbb{1}_{[0,+\infty)}(z)$	m	$2m$	$\frac{1}{(1-2t)^{\frac{m}{2}}}$ per $t < \frac{1}{2}$
$\mathcal{W}(\alpha,\lambda)$ (Weibull)	$\alpha, \lambda > 0$	$\alpha\lambda\,z^{\alpha-1}\,\mathrm{e}^{-\lambda z^\alpha}\,\mathbb{1}_{[0,+\infty)}(z)$	$\frac{\Gamma\left(1+\frac{1}{\alpha}\right)}{\lambda^{\frac{1}{\alpha}}}$	$\frac{\Gamma\left(1+\frac{2}{\alpha}\right) - \Gamma\left(1+\frac{1}{\alpha}\right)^2}{\lambda^{\frac{2}{\alpha}}}$	–
$N(\mu,\sigma^2)$ (normale)	$\mu \in \mathbb{R}$ $\sigma^2 > 0$	$\frac{1}{\sigma\sqrt{2\pi}}\,\mathrm{e}^{-\frac{(z-\mu)^2}{2\sigma^2}}$	μ	σ^2	$\mathrm{e}^{\frac{\sigma^2 t^2}{2}+\mu t}$
$LN(\mu,\sigma^2)$ (log-normale)	$\mu \in \mathbb{R}$ $\sigma^2 > 0$	$\frac{1}{z\sigma\sqrt{2\pi}}\,\mathrm{e}^{-\frac{(\ln z-\mu)^2}{2\sigma^2}}\,\mathbb{1}_{(0,+\infty)}(z)$	$\mathrm{e}^{\frac{\sigma^2}{2}+\mu}$	$\left(\mathrm{e}^{\sigma^2}-1\right)\mathrm{e}^{\sigma^2+2\mu}$	–
$t(m)$ (t di Student)	$m \in \{1, 2, \ldots\}$	$\frac{\Gamma\left(\frac{m+1}{2}\right)}{\sqrt{\pi m}\,\Gamma\left(\frac{m}{2}\right)}\left(1+\frac{z^2}{m}\right)^{-\frac{m+1}{2}}$	0 per $m > 1$	$\frac{m}{m-2}$ per $m > 2$	–
$F(m,n)$ (F di Fisher)	$m, n \in \{1, 2, \ldots\}$	$\frac{\left(\frac{m}{n}\right)^{\frac{m}{2}}\Gamma\left(\frac{m+n}{2}\right)\,z^{\frac{m}{2}-1}}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)\left(1+\frac{m}{n}\,z\right)^{\frac{m+n}{2}}} \times \mathbb{1}_{(0,+\infty)}(z)$	$\frac{n}{n-2}$ per $n > 2$	$\frac{2n^2(m+n-2)}{m(n-2)^2(n-4)}$ per $n > 4$	–

Densità di vettori aleatori assolutamente continui

$(X,Y) \sim$	Parametri	$f_{(X,Y)}(u,v)$
$\mathcal{U}(D)$ (uniforme sul piano)	$D \in \mathbb{R}^2$	$\frac{1}{\mathrm{area}(D)}\,\mathbb{1}_D(u,v)$
$N(\mu_X,\mu_Y,\sigma_X^2,\sigma_Y^2,\rho_{X,Y})$ (normale bivariata)	$\mu_X, \mu_Y \in \mathbb{R}$ $\sigma_X^2, \sigma_Y^2 > 0$ $\rho_{X,Y} \in (0,1)$	$\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{X,Y}^2}}\,\exp\left\{-\frac{1}{2(1-\rho_{X,Y}^2)}\left[\left(\frac{u-\mu_X}{\sigma_X}\right)^2+\left(\frac{v-\mu_Y}{\sigma_Y}\right)^2 - 2\rho_{X,Y}\left(\frac{u-\mu_X}{\sigma_X}\right)\left(\frac{v-\mu_Y}{\sigma_Y}\right)\right]\right\}$

Nota: Se $(X,Y) \sim N(\mu_X,\mu_Y,\sigma_X^2,\sigma_Y^2,\rho_{X,Y})$, allora

$$\mathbb{E}[X] = \mu_X\,, \qquad \mathbb{E}[Y] = \mu_Y\,, \qquad \mathrm{Var}[X] = \sigma_X^2\,, \qquad \mathrm{Var}[Y] = \sigma_Y^2\,, \qquad \mathrm{Cov}[X,Y] = \rho_{X,Y}\,\sigma_X\,\sigma_Y$$

$$f_{Y|X}(v \mid u) = \frac{1}{\sigma_Y\sqrt{2\pi(1-\rho_{X,Y}^2)}}\,\exp\left\{-\frac{1}{2\sigma_Y^2(1-\rho_{X,Y}^2)}\left[v-\left(\rho_{X,Y}\,\frac{\sigma_Y}{\sigma_X}\,(u-\mu_X)+\mu_Y\right)\right]^2\right\}$$

Statistiche notevoli

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i\,, \qquad S_n^2 = \frac{1}{n-1}\sum_{i=1}^n \left(X_i - \bar{X}_n\right)^2, \qquad S_{0,n}^2 = \frac{1}{n}\sum_{i=1}^n \left(X_i - \mu\right)^2, \qquad S_p^2 = \frac{(m-1)\,S_{X;m}^2 + (n-1)\,S_{Y;n}^2}{m+n-2}$$